# Chapter 2_The 5 level of the Agentic AI

SPAR (Sense, Plan, Act, Reflect) is one of most easiest analogy to understand the agentic AI.

- **Sense:** the agent maintain the 'short-term context window' similar to how a self-learning car keeps track of immediate road conditions and navigation requirement.
- **Plan:** the ai agent engage in sophisticated reasoning to develop the step-by-step plans for achieving their goals.
- **Act:** Agents use their available tools to carry out actions in their environment, whether that's sending communications, updating systems, or managing the digital resources.
- **Reflect:** AI agent can evaluate their performance, analyze the outcomes, and refine their approaches based on what works best.

## The 5 levels of the AI Agent

| Level | Car Analogy | Agentic AI Analogy | Main Technology Involved | SPAR Capabilities (Sensing, Planning, Acting, Reflecting) |
| --- | --- | --- | --- | --- |
| Level 0 - Manual Operations (Human-Only) | Manual driving with no assistance. | Humans perform all tasks without automation. | Basic digital tools (spreadsheets, email), manual processing. | NA |
| Level 1 - Rule-Based Automation | Basic cruise control maintains speed but needs human operation. | Simple automation follows fixed rules (e.g., data entry, RPA systems). | Basic automation tools (RPA, simple scripts, rule engines). | ***Sensing:*** Predefined triggers and structured data.<br>***Planning:*** Simple if-then rules and decision trees.<br>***Acting:*** Deterministic actions based on fixed inputs.<br>***Reflecting:*** No true learning, only logging and error reporting. |
| Level 2 - Intelligent Process Automation | Advanced driver assistance systems handle speed and steering with supervision. | AI combines automation with cognitive abilities like NLP and machine learning. | AI tools (machine learning, NLP, computer vision, RPA, process orchestration). | ***Sensing:*** Semi-structured data from multiple sources.<br>***Planning:*** Basic AI models for pattern recognition and decision-making.<br>***Acting:*** Sophisticated actions with error handling.<br>***Reflecting:*** Basic analytics and performance monitoring, no adaptive capabilities. |
| Level 3 - Agentic Workflows | Vehicles navigate highways but need human intervention in complex situations. | Agents generate content, plan, reason, and adapt in defined domains. | Large language models, memory systems, content generation tools, basic reinforcement learning. | ***Sensing:*** Advanced natural language understanding and context awareness.<br>***Planning:*** Reasoning using foundation models, orchestrating complex workflows.<br>***Acting:*** Chaining tools and handling multi-step tasks.<br>***Reflecting:*** Limited short-term feedback adjustments and long term memory. |

| Level | Car Analogy | Agentic AI Analogy | Main Technology Involved | SPAR Capabilities (Sensing, Planning, Acting, Reflecting) |
|---|---|---|---|---|
| Level 4 - Semi-Autonomous Agents | Self-driving cars operate autonomously in specific conditions. | Agents work autonomously within defined expertise, adapt strategies, and learn. | Advanced reasoning and planning, real-time adaptation, causal reasoning. | ***Sensing:*** Multi-modal perception and interpretation of diverse inputs.<br>***Planning:*** Dynamic strategies for complex tasks and goal breakdown.<br>Acting: Autonomous tool usage and error recovery.<br>***Reflecting:*** Retains context across sessions, learns from past experiences. |
| Level 5 - Fully Autonomous Agents | Fully autonomous cars drive anywhere in all conditions. | AI systems handle any task, cross-domain learning, and self-adaptation with no human intervention. | Sophisticated memory systems, advanced learning mechanisms, safety protocols for autonomy. | ***Sensing:*** Complete environmental awareness and goal formulation.<br>***Planning:*** Advanced reasoning and original problem-solving.<br>***Acting:*** Full autonomy in tool selection and execution.<br>***Reflecting:*** Continuous self-improvement, robust long-term memory. |

Table 1.3: The Agentic AI Progression Framework (Source: © Bornet et al.)



Now, most AI agents on the market operate at level 2 or 3, with some specialized systems reaching level 4 in narrow domains. And the level 5 is the future goal.

**Golden rule:** the simpler, the better. and the best "level" depends entirely on your specific needs and circumstances.

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'Autonomy increases while direct human control decreases'
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